Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Is Statistics and Why Do We Need It?

Every day we come across lots of information. The number of students in your class, how many bottles of water were sold at a shop in a week, the heights of your friends — these are all pieces of information. When we collect such facts together for a particular reason, we call them data. Data are facts or information collected for a particular purpose.

Statistics is the science of collecting, organising, presenting, analysing and interpreting data. So statistics is not just about gathering numbers — it is the whole way of handling information so that it becomes useful. Notice the five actions in that definition:

Collecting — gathering the information you need.
Organising — putting that information in order so it is easy to work with.
Presenting — showing the information clearly, for example in a table.
Analysing — studying the information to see what it tells you.
Interpreting — explaining what the information means.

Why do we need statistics? It helps us to:

Collect information — it shows us how to gather facts in an orderly way.
Organise information — it lets us arrange the facts neatly so nothing is lost.
Compare information — it lets us look at two sets of facts and see which is bigger, faster or more.
Make decisions — with facts organised, we can choose wisely instead of guessing.
Present information clearly — it lets us show others our findings in a way they can understand.

Worked example — seeing statistics in action

Suppose your class wants to know how many students come to school by bus, by bicycle and on foot. Here is how statistics helps at each step:

Collect: You ask every student in the class how they travel and write down each answer.
Organise: You group the answers — all the "bus" answers together, all the "bicycle" answers together, all the "foot" answers together.
Present: You put the totals in a table so anyone can read them at a glance.
Analyse and interpret: You look at the table and see that most students come by bus, so you can explain that to your class.

That last point is an example of a decision: seeing that most students travel by bus, your class might arrange a bus pick-up point. Statistics turned scattered answers into something useful.

Practice

1. In your own words, what are data? Give one example of data you might collect at home.
2. Define statistics by naming the five actions it involves.
3. List three ways statistics helps us. For each one, say in a sentence how it is useful.

4. A shop owner writes down how many loaves of bread she sells each day for a week. Write the five steps of statistics in the order she would use them, from collecting the data to explaining what the data mean.

Section 2: Collecting, Recording and Tabulating Data

Statistics is the study of information — called data — gathered about people, things or events. Before we can read and understand a table of numbers, the data must first be collected, and then set out neatly. There are four common ways to collect data:

1. Observation — you watch something happen and note what you see.
2. Counting — you count how many of each item there are.
3. Interview — you ask people questions face to face and record their answers.
4. Questionnaire — you give people a set of written questions to answer.

Whichever method you use, you end up with a list of items. For example, counting the vehicles passing a school might give you this list: bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car. A list like this is hard to read, so we use tally marks to record the data as it is collected.

Tally marks are short strokes used to record each item as it is counted. Every time you see one more of an item, you add a stroke. Strokes are grouped in fives: four upright strokes, then a fifth stroke drawn across them like ||||/. This grouping makes it easy to count the total at a glance.

Let's record the vehicle data with tally marks. The data is: bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car.

- car appears 6 times → tally ||||/ |, total 6
- bus appears 1 time → tally |, total 1
- lorry appears 2 times → tally ||, total 2
- bicycle appears 3 times → tally |||, total 3

When the tally is finished, you can count up the strokes for each item and write the total in the last column. A table that shows how many times each item occurs in a set of data is called a frequency table. In a frequency table, the last column is the frequency — the number of times each item occurs.

Let's set the vehicle data out as a complete frequency table. First list each different vehicle in its own row, record the tally marks as you count, then write the total.

Vehicles	Tally	Total
car	||||/ |	6
bus	|	1
lorry	||	2
bicycle	|||	3

The frequency table shows at a glance that car is the most common vehicle (6), and bus the least common (1).

A frequency table does not always need a tally column — sometimes it just shows each item and its frequency. For example, a survey of students' favourite colours gives this frequency table:

colour	frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

Here the most popular colour is Pink (30 students) and the least popular is Green (14 students). By setting data out in a table like this, you can read it quickly and answer questions about it easily.

Practice

1. State three methods of collecting data.
2. Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
3. The favourite colours of students are recorded with these frequencies: Red 18, Blue 20, Green 14, Yellow 16, White 15, Pink 30. Which colour is the most popular and which is the least popular?

Check your work

Attempt every question before you read this page.

What Is Statistics and Why Do We Need It?

1. Data are facts or information collected for a particular purpose. An example could be the number of plates washed each evening, or the amount of time spent on homework each day — any facts you collect for a purpose.
2. Statistics is the science of collecting, organising, presenting, analysing and interpreting data.
3. Three of: it helps us collect information (gather facts in an orderly way), organise information (arrange facts neatly), compare information (see which set is bigger or more), make decisions (choose wisely instead of guessing), and present information clearly (show others our findings).
4. Collect (write down the daily sales), organise (arrange the figures, for example from smallest to largest or by day), present (show the figures clearly, for example in a table), analyse (study the figures to see patterns), interpret (explain what the figures mean, such as which day sells most bread).

   The definition of statistics lists the five actions in order: collecting, organising, presenting, analysing, then interpreting. So the shop owner first collects her daily sales figures, then organises them neatly, then presents them clearly in a table, then analyses them to spot patterns, and finally interprets them — explaining what they tell her, such as the busiest day.

Collecting, Recording and Tabulating Data

1. Observation, counting, interview and questionnaire (any three of these).
2. | Number | Frequency |
   |---|---|
   | 1 | 3 |
   | 2 | 3 |
   | 3 | 4 |
   | 4 | 3 |

   Count how many times each number appears. The number 1 appears at positions 1, 2 and 12 → 3 times. The number 2 appears at positions 7, 8 and 13 → 3 times. The number 3 appears at positions 5, 6, 9 and 10 → 4 times. The number 4 appears at positions 4, 5 and 11 → 3 times. Set these counts in a table with a Number column and a Frequency column.
3. Pink is the most popular (30), and Green is the least popular (14).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-18e3e608	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____